UNITED NATIONS DEVELOPMENT PROGRAMME

PROJECT DOCUMENT

Sudan

Empowered lives.
Resilient nations.

Project Title: Solar water pumps for Sustainable Agriculture in Sudan: Promotion of renewable energy and sustainable financial modality as gateway to improve farmers' livelihoods and GHG reduction

Project Number: Award ID – 00106169; Output ID – 00107041

Implementing Partner: United Nations Development Programme

Start Date: 1-4-2019 **End Date:** 31-3-2023 **PAC Meeting date:** 5-2-2018

Brief Description

The project aims to support sustainable agriculture in Sudan by promoting the use of solar photovoltaic (PV) instead of diesel-based irrigation water pumping in farmlands. The adoption of solar PV technology for water pumping in agriculture avoid negative impact on air pollution and dependence on fossil fuels which are imported.

The project comprises three outcomes: install solar pumps in the farmlands and an accompanying financing mechanism for continued finance of project; technical know-how on effective management of the agronomic practices and water in the farmland; and promote enabling activities to access the fund for scaling up and further replication of the solar pumps.

According to the Sudan's Cabinet approval, resolutions number 58/2017, the Government of Sudan has endorsed the exemption of all solar powered irrigation equipment from customs and other fees (letter attached in annex). At the same time, the Ministry of Finance and Economic Planning has approved a national PV fund (see annex 7) with the aim of promoting the use of solar energy operated water pumps for irrigation in Sudan. The formation and formulation of rules and regulations for National PV fund is ongoing and it is expected to be materialized by the end of 2018. The project will will install 450 solar pumps with the KOICA fund and the farmers will reimburse the initial investment of the solar pump according to the rules and regulations of the financing mechanism (National PV fund). The KOICA fund will be used as the seed funding for the National PV fund.

Contributing Outcome (UNDAF/CPD) By 2021, people's resilience to consequences of climate change, environmental stresses and natural hazards is enhanced through strengthened institutions, policies, plans and programmes
Indicative Output(s) with gender marker[2]:

Total resources required:	USD 7,000,000	
Total resources allocated:		
	UNDP TRAC:	600,000
	Donor:	6,400,000
	Government In-Kind:	Office for the project Implementation Unit and assignment of a Government project Coordinator
Unfunded:		

Agreed by (signatures)[1]:

	Government In-Kind:	Office for the project Implementation Unit and assignment of a Government project Coordinator
Unfunded:		

Agreed by (signatures)[1]:

Government	UNDP	Implementing Partner
Dr. Abdel`moneim Mohamed Eltayeb Undersecretary- Ministry of Finance and Economic Planning	Dr. Selva Ramachandran Resident Representative	Eng. Mohamed Khair Elginaid Undersecretary – Ministry of Water Resources, Irrigation and Electricity
Date: 21-5-2019	Date: 21-5-2019	Date: 21-5-2019

[1] Note: Adjust signatures as needed

[2] The Gender Marker measures how much a project invests in gender equality and women's empowerment. Select one for each output: GEN3 (Gender equality as a principle objective); GEN2 (Gender equality as a significant objective); GEN1 (Limited contribution to gender equality); GEN0 (No contribution to gender quality)

I. DEVELOPMENT CHALLENGE

Traditional rain fed agriculture is the predominant production system in rural Sudan. It covers about 9.0 million hectares, representing more than 50% of the total national cultivated land, and represents the major livelihood source for the majority of the population. Crop production is entirely dependent on rainfall, small scale irrigation by using diesel water pumps and is typically small-scale and subsistence-oriented. In River Nile state, irrigation is concentrated mainly around the River Atbra while rain fed agriculture is practiced in other parts of the state remote from seasonal streams. Yields are very low due to environmental conditions and recurrent droughts. Traditional rain fed agriculture has become excessively risky and uncertain due to increasing climatic variability and competitive pressures on natural resources. The farmers also use irrigation pumps which is along the banks of the River Atbara and most of which are private holdings, small in farm size ranging between less than 1 to 10 hectares. Diverse types and sizes of pump-sets to lift water to the level of farm. The main crops are wheat, beans, dura, spices, vegetables and diverse types of fruit tress (citrus and date palm). Diesel pumps are the commonly practiced types for pumping water for irrigation. However, currently this practice is facing serious problems of diesel shortage and escalating prices of diesel fuel which make the agriculture production is unprofitable and threaten the livelihoods of farmers in those areas irrigated with diesel pumps. On the other hand, small scale irrigation, by using diesel water pumps, are facing issues such as the high cost of diesel fuel, and the time spent in searching for diesel and transportation to the farm (around 8 hours) and anxiety of not securing the diesel on time.

Emerging climatic trends such as declining precipitation, more frequent drought episodes, higher temperatures, more intense sandstorms are leading to a sharp degradation of natural resources and exposing farmers to higher risks. Compounding this situation are factors such as poor infrastructure including limited access to irrigated water, weak access to inputs/services, low technical capacity, and chronic poverty conditions. Farmer households remain highly vulnerable to these intensifying threats to the productivity of subsistence agricultural systems. Recent efforts to improve resilience to current climate variability and future climate change have not acceptably lowered vulnerability. Hence, there is need for implementing cost-effective mitigation and adaptation strategies to improve yields, promote livelihood safety and reduce income volatility in the face of climate change in Sudan including River Nile State. Therefore, the project will focus on switching the diesel pumps with solar pumps that the participating farmers will enjoy the benefit from saving the cost of diesel, saving the time spent in searching for available diesel and transportation to the farm, being relieved from the anxiety of not securing the diesel on time, expanding the cultivated land, choosing more profitable crops, reducing the GHG emission and possible to participate GHG emission rights exportable to Korea, EU or other potential countries. By switching the diesel pumps with PV pumps and providing farmers with training for better management of water, soil and crop choices, the farmers could improve crop productivity resulting to increase their net agriculture income, making their agriculture practice more sustainable and contributing to reducing the climate change.
The key objective of the proposed project is to establish a good pilot model of replacing the diesel pumps with PV pumps that can be replicated in River Nile State and in other states later on. In the process of this pilot project, a strong partnership with the federal and state government is important both for the success of the pilot project itself and the successful scaling up of the pilot project by the federal and state government.

II. STRATEGY

River Nile state is in the semi-desert area of Sudan and irrigated agriculture is practiced widely in the State; the total irrigated are is estimated as 150,000 hectares. Common crops grown are wheat and faba beans which are cultivated in an area of 31,700 hectares 25,000 hectares respectively and represents the main cash crops. Other crops cultivated include sorghum chickpea, dry beans, onions, potato, vegetables and different spices.

The agricultural sector is the largest share of CO_2 emissions in Sudan among the other sectors including forest, industrial, waste and energy. In 2000, the emissions from agriculture sector,

57,611 Gt of CO_2, represented approximately 74% of the total emissions produced in the country.[2] The GHG emissions as reported are mainly from enteric fermentation and manure management with no GHG emissions attributed to the use of diesel for pumping water.

Sudan is largely dependent on imported fossil fuels. Hence, there is an urgency to implement Sudan's Renewable Energy Master Plan (REMP) and reduce Sudan's dependence on fossil fuel. Sudan has abundant solar resources but largely lacks the capacity to utilize solar resource for power generation. This project will contribute to the implementation of REMP.

Approximately 35% of Sudan's population has access to electricity.[i] In 2012, the power consumption per capita was 233 kWh/ year.[ii] There are no independent power producers (IPPs) in the country, though initiatives are underway to promote private investment in power generation.[iii] The majority of Sudan's land areas receive solar irradiation greater than 2,300 kWh/m^2/year, which is considered an excellent solar resource. To take advantage of Sudan's plentiful solar resources, this project targets off-grid areas in River Nile State.

This proposed project will support the implementation of Sudan's Twenty-Five Year National Strategy (2007–2031); Renewable Energy Master Plan Plan 2005, Five-Year Programme for Economic Reform (2015-2019); Highlights of the Four-Year Development Plan 2017–2020; Interim Poverty Reduction Strategy Paper (2011-2016); National Agriculture Investment Plan (2016–2020); National Adaptation Programme and Action 2007, National Communications Report 2013, National Adaptation Plan, 2014 and Intended National Determined Contribution Report INDC 2015.

The project is consistent with Sudan's national strategies, as evidenced by the already-existing incentives for renewable power in Sudan's Investment Act. The project will help further the goals Sudan's national strategies by putting in place the overall framework that will make them effective. The Technology Needs Assessment (TNA) carried out by HCENR with KOICA support points to renewable energy as one of Sudan's key priorities in climate change mitigation. Similarly, Sudan's Second National Communication 2013 to the UNFCCC includes renewable energy as "a key potential mitigation option". The Sudanese government's Five-Year Programme for Economic Reform (2015-2019) also makes strong references to achieving the SDGs in Sudan. The Five-Year Plan aims to achieve economic stability and the leap in all economic fields, increasing production and productivity and providing employment opportunities by utilizing resources (human and capital) and economic partnerships opportunities with brother economic partners where improving the infrastructure is key, e.g. water and electricity sectors to implement efforts in the agriculture and manufacturing sectors.

The project is also in-line with the Interim Poverty Reduction Strategy Paper (IPRSP, 2011) which emphasizes the promotion of economic growth and employment creation through agriculture sector as the first pillar of the Government of Sudan's development strategy. The agriculture sector is one of the three priority sectors mentioned in the NAPA 2007 and the NAP 2014. The sectors include water, agriculture (both livestock and crop production systems), coastal zone and human health. The INDC 2015 recognizes solar pumping in agriculture sector, and on basis of vulnerability assessments done under the NAP (2014), proposes activities for adaptation.
The proposed Project responds directly to the NAPA and NAP, addresses several of the highest NAPA and NAP priorities. Identified among adaptation needs according to NAPA, section 3 - Identification of Adaptation Needs is enabling the introduction to micro-credit to support adaptation activities. The project will support NAPA priorities through the provision of micro-finance services that are better tailored to addressing climate risks and other innovative risk finance instruments, such as loan for solar pumps, to help farmers to better manage covariate risk in agriculture sector. According to the River Nile State Adaptation Plan (NAP), one of the key priorities area is solar pumping for agriculture and will contribute SDG1: Eradicate Extreme Poverty; SDG2: Zero Hunger; SDG5: Gender Equality; SDG7: Affordable and Clean Energy and SDG13: Climate Action.

[2] Sudan's Second National Communication under the United Nations Framework Convention on Climate Change, January 2013.

The project interventions are in line with the following strategic plan and actions.

A. In line with KOICA's mid-term strategy (2016-20)
The project contributes to KOICA's SO1 "Promote Sustainable Development through Climate Change Response" and its outcome number 9.1.1 (reduction in greenhouse gases), 9.1.2 (improvement in climate adaptive capacity) and 9.1.3 (mainstreaming of climate projects). The project promotes "Investment in Renewable Energy Technologies", recognizing that renewable energy plays an indispensable role not only in combating global climate change but also in addressing energy access, energy security, environmental pollution and sustainable development. In accordance with the adopted strategy, the KOICA support goes beyond the creation of a financing scheme and promotes direct investment as well as mechanistic changes which will promote the adoption of solar water pumping.

The specific outcomes of the KOICA's mid-term strategy 2016-20 that the project will address:

- Reduction in Greenhouse Gases (KOICA's outcome 9.1.1)
- Improvement in climate adaptive capacity and (KOICA's outcome 9.1.2)
- Mainstreaming climate adaptive measures and solar water pumping activities in regional development and sectorial plan (KOICA's outcome 9.1.3)

B. The project is consistent with Sudan's national strategies

The project aims to develop and accelerate the adoption of off-grid solar pumping by providing demonstration units, technical capacity building, a quality assurance mechanism, and a financing mechanism. Sudan's development depends critically on the ability to reduce dependence on fossil fuels as Sudan is currently a fossil fuel-poor nation. The present lack of availability of alternatives to diesel pumping constrains farmers, often limiting the plot of the land which they can plant. Lack of locally tested and trusted technological alternative, lack of user experience with the technology, and lack of financing, has meant that adoption of alternatives to diesel pumping in off-grid areas is slow and limited.

The project will play a critical role in strengthening a market which is pre-matured stage and supporting it through pre-mature stage to the point where it is self-sustaining and able to respond to the needs of the farmers.

C. Country ownership
The objective of the project is consistent with the strategies of the Sudanese Government's strategies, particularly as outlined in the Renewable Energy Master Plan (2005). Sudan has been initiated the development of a Nationally Appropriate Mitigation Actions (NAMA) to support renewable energy in 2015. As a result, this projectwill link with NAMA and support to secure international carbon finance to support the long-term development of solar pumping in Sudan. It will thus provide Sudan with the opportunity to reinforce its engagement with the international climate change architecture and demonstrate its commitment to international efforts to reduce GHG emissions.

D. Cost-effectiveness
The KOICA financing for Outcome 1 (US$ 4,065,900), represents the bulk of the KOICA financing for the project and has been allocated to support the development of pilot solar PV projects. These are seen as the most critical step in launching solar pumping in Sudan by demonstrating to farmers that solar pumping is viable and demonstrating to bankers that it is a reliable, financeable activity. The success of these solar pumping demonstrations will translate to future projects while a failure will setback solar pumping in Sudan by several years.

At present, no entity is willing or capable of putting forth the finance and technical support necessary for such a demonstration. Hence, UNDP-KOICA support will be critical in implementing these demonstration systems and doing it in a way that can prove successful and inspire the confidence of future stakeholders. The KOICA investment of $4,065,900 in this component will in-turn result in annual savings of 2,970 tons of diesel fuel; which is equivalent to $ 775,500 per year

will be retained by farmers once they have paid off the value of their pumps. The detailed calculation is attached in annex.

The KOICA financing for Outcome 2 ($509,250) assures cost-effectiveness in two principal ways. First, it will serve to guarantee the quality of the ($3,717,900) worth of pumps purchased under the project and that they are suitably sized and selected for the conditions of their application. Second, the water efficiency component will ensure that the amounts of water needed are optimized and therefore the pump size, and associated capital cost can be minimized for a given crop and area.
The KOICA financing for Outcome 3 ($576,000) consists of technical assistance to enable documentation, solar pumps activities are integrated into state development plan and dissemination of experience gained in the present project in the River Nile State for replication in other areas.

The proposed project is extremely cost-effective as it will utilize relatively limited KOICA funds to leverage investments in agriculture throughout Sudan. The potential for replication in Sudan and other areas is significant. Water pumping is problematic and costly in most of Africa and relies on imported, hard to obtain, diesel. With a demonstrated alternative, adoption can be expected to spread quickly. The cost effectiveness of the project is reflected in its GHG abatement cost of $19.56/ tCO_2 of direct emissions.

The process has been validated by extensive stakeholder consultations as effective to minimize and reverse the impact of weather and climate impacts on small and medium farmers, thereby reducing vulnerability of rural communities from increasing climatic variability and climate change. In general terms, these initiatives involve a hierarchy of activities that span risk minimization, risk absorption, and risk avoidance strategies in water and agricultural systems. Therefore, the objective of the project is to enhance the sustainable agriculture in Sudan by promoting the use of solar photovoltaic (PV) instead of diesel-based irrigation water pumping in farmlands. Together, these activities comprise a set of specific measures that span three major outcomes: to install solar pumps in the farmlands and an accompanying financing mechanism for continued finance of project; technical know-how on effective management of the water in the farmland; and promote enabling activities to access the fund for scaling up and further replication of the solar pumps.

The **first outcome** is concerned with installing of solar pumps. The project will conduct baseline survey in the initial period of the project life and depending on the analytical results of the baseline survey, the number of total PV pumps and the composition of small pumps and medium pumps are subject to change. However, based on the available information, the project is expected to install 450 pumps (250 ×3.12 kWp units and 200×5.12 kWp units) with the support of KOICA fund and farmers will reimburse initial investment of the solar pump cost according to the rules and regulations of the financing mechanism (National PV fund). To be able to achieve the result of outcome, the project will i) provide a technical and financial model of effectively implementing the water pump replacements augmented with training of farmers for more efficient management of water, soil and crop choices and ii) KOICA fund will be used as the seed funding for the National PV fund once established. The focus of these activities is the dissemination of PV pumps in the River Nile State.

In the River Nile State, the prevailing practice of access to water for farmland is the use of diesel engines to pump water in areas which are not connected to the grid. This outcome promotes the adoption of PV pumps in two ways:

a. Providing installation of 450 pumps (250×3.12 kWp units, 200×5.12 kWp units) to serve as a demonstration and to facilitate gathering of data on the operation of the pumps; and
b. Creating a River Nile State PV fund and link with National PV fund that provides a finance mechanism which removes the barrier of the large capital required for replication of solar pumps in River Nile State. The Ministry of Finance and Economic Planning has endorsed to form National PV fund in July 2017 (letter attached in annex). At the same time, the Cabinet of Sudan has been approved tax exemption of all solar powered irrigation equipment from customs and other fees (letter attached in annex). The National PV fund is in the process of establishment (in base line GEF funded project) in consultation with potential participating Banks at headquarters level in Khartoum and then transferred for execution in the River Nile

State. At the National level, it is expected that a steering committee will be in place in 2018, since the cabinet and the Ministry of Finance approved to establish the National PV Fund. The steering committee would consist of participating commercial bank representatives, the Ministry of Finance and other relevant government entities and beneficiary communities. The project management team will be equipped with a legal advisor, technical, administrative and financial teams and coordinates with the banks consortium at headquarters in Khartoum. At the state level, the project will ensure that the State Ministry of Agriculture (supported with technical, administrative and financial teams), coordinates with the bank consortium at branches level. The seed money for the National PV fund will be provides as follows. Initially, the cost of the installing the solar pumps will be paid with the KOICA fund and then 80% of the installation cost and interest will be paid back by the farmers on instalments according to the terms specified by the financial arrangement. The flow of the farmers' payments will be tapped into the National PV Fund as seed money. The seed money will be provided on the condition that the interest rate and the repayment schedule of the National PV Fund are set in favour of the farmers. (i.e. the annual interest rate is approximately the range of 9~12%, and the farmers start pay back after the first harvest of the major crops since the installation of the PV pump)

Outcome 1: Financing and dissemination mechanism established and operational to support a PV pump installation programme *(KOICA Outcome 9.1.1. Reduction in greenhouse gases – indicator: the amount of CO2 reduced (tons CO2 per eq)*

Output 1.1: 450 pumps installed as part of a demonstration phase with KOICA fund
Output 1.2: Prepared a pilot financial scheme that the River Nile State PV fund can refer to and the seed money on the condition that the financial arrangement of the fund is consistent with the pilot financial model

450 demonstration pumps are expected to be installed within 30 months of project start. During the first year, the project will set up River Nile State PV fund with the support of technical guidance and advice by the project for setting up the fund and link with National PV Fund which is expected to be established by the end of 2018. The project will prepare a financial scheme that the River Nile State PV fund can refer to and the seed money on the condition that the financial arrangement of the fund is consistent with the pilot financial model. The data collected and the efforts to establish the River Nile PV fund and link with National PV fund aiming to subsidy level to be offered, the loan term and cost of finance. At the state level, the project will ensure that the State Ministry of Agriculture (supported with technical, administrative and financial teams), coordinates with the bank consortium at branches level and make sure for repayment to the PV fund as per terms and conditions of the agreement. The Idea is that farmers will start to pay after one year of installation and the whole instalment will fully be paid after the project completion as the calculated payback period for small farm is 5.06 years and that for medium farm is 4.58 years. Farmers will start to pay in a time of 6 months before the project ends but will continue to pay their debts to either State PV fund or National PV fund which ever in place at that time. The money will be reinvested for more solar pumps for other farmers

The **second outcome** aims to promote technical know-how on agronomic practices and increase efficiency of water use, thereby increasing the overall sustainability of pumping practices and reducing the size (and therefore cost) of solar PV pumps and combination with improved practices for effective and efficient management of water, soil and crop choice by the farmers. the project will formulate training manuals on agronomic practices and water efficiency, training of the Sudanese experts in Korea and the training sessions for farmers by the Agriculture Research Center. The project will focus on sustainable water management practices such water harvesting and efficient use of water for irrigation that increase water storage in soil and groundwater and at homesteads and reduce water losses leading to improve the crop yield.

When using a diesel pump, if a farmer desires more water it is possible to operate for longer hours. When using a solar pump, the pump must be sized in advance for a certain capacity. The size of the pump determines its cost, with the relationship between size and cost being almost linear. A reduction in the amount of water required translate directly to a reduction in the capacity of the PV system, and therefore directly to a reduction in the capital cost required. Thus, the

efficient use of water has the potential to make solar PV pumps considerably more economical while reducing the impact on underground water sources.

Outcome 2: Improved in climate adaptive agronomic practices and efficiency of water usage in the farm land *(KOICA Outcome 5.2.1. Improved in efficiency of water usage; Indicator: Increase volume of water availability; Increase of irrigation area – hectare; Outcome 9.1.2. Improvement in climate adaptive capacity – indicator: the increase rate of major crops production %)*

Output 2.1: Enhanced farmers and agriculture extension staff member's skills and knowledge in soil, crop, agronomic practice and water management to undertake sustainable farm management
Output 2.2: Improved efficiency of water usage by using different types of small scale irrigation methods, management and maintenance of the irrigation infrastructure and solar pumping stations delivered to benefit 450 farmers

To achieve the strategy on efficient use of water in farmland, the KOICA fund will primarily through the use of:

a. Support structures with the active participation of farmers, develop training manuals, training sessions for farmers by ARC and training of Sudanese experts in Korea and
b. Combined with agronomic practices

The fund will support for soil conservation and water harvesting structure (typically water distribution system, stone, earth or vegetative cross-slope barriers, bunds, tied ridges, drip irrigation, canal line up, terraces and proper levelling) for water management and efficiency. On the other hand, in-field practices (including mulching, fertilizing, intercropping, crop rotation, agroforestry and reduced tillage) will conduct for improvement of the water efficiency. Agriculture Research Center will support technical assistance for structure development (i.e. small-scale irrigation technique, soil conservation, management and maintenance of the irrigation structure and training about cropping pattern) for the water efficiency and agronomic practices based on the locally adapted technologies. In addition, the project will develop training manuals on soil conservation and water harvesting techniques, integrated farm management and maintenance on solar system maintenance. Moreover, the project will conduct training of Sudanese experts in Korea. ARC will train to the extension staff from ministry of agriculture and farmer leaders. The replication of technical know-how scheme will conduct through ToT approach to other farmers.

The **third outcome** supports the scaling-up and expansion of the project to other states in Sudan. It includes a structured replication programme and knowledge sharing programme to replicate success in the River Nile to other States of Sudan.
One of the key activities of the previous Electric Water Pump project, the Ministry of Finance and the Ministry of Agriculture to include PV pumps in the fiscal concessions list of the Investment Law and the Agricultural Implements Regulation. The fiscal concessions granted by the Investment Law and the exemption from taxes and duties once PV pumps are classified as 'agricultural equipment' will serve to lower hardware prices and will benefit consumers (i.e. farmers), equipment suppliers/retailers seeking to grow the market, and banks providing finance to farmers (shortening loan repayment times and reducing risk exposure). Fiscal benefit will allow more (cheaper) pumps to be subsidized and hence increase the emissions reduction impact of the project. However, it is not enough to take up of the PV to other states and therefore, the KOICA fund will use for structured replication programme for other states and codified, documented and share best practices (type of solar water pumps and adaptation measures) to and integrated into the state level and national policies. The concessions will have national force and will, therefore, also promote nationwide take-up of PV pumps with the support of KOICA fund.

Outcome 3: Supportive enabling environment and scaled-up implementation
Output 3.1: Successfully demonstrated practices (type of solar water pumps and adaptation measures in farmland) codified and documented
Output 3.2: Designed structured replication programme for other states (at least two states) and integrated into the state level/or agriculture sector development plan

In supporting the scaling up and expansion, the project will develop manual of implementation including effective means of advertising the project (i.e. Information, Education and

Communication – IEC materials), soliciting applications, the criteria of selection for targeted farmers which will be derived from the analysis of the baseline survey and detailed instructions on selection process, installation protocol (i.e. requirements for equipment and installation including various gauges and flood proof height of solar panel), training for the use and maintenance of PV pumps and procedures for monitoring the farming practices and agriculture income. Replication will be encouraged in other states and areas by demonstrating, documenting, and disseminating the avoided costs of infrastructure and fossil fuels, as well as the environmental benefits of avoiding diesel use, thereby incentivizing governments in other states to adopt measures, outlined under output 3.2 to promote solar water pumping in place of diesel.

The Project will coordinate between the River State Government, and the commercial banks to develop a customized lending mechanism for solar water pumping systems based on experiences of previous pilot project. The KOICA project, together with the River Nile State Government and other stakeholders will provide a series of initiatives to ensure the success of the project under the outcome 1 during the project life time. The outputs and the detailed activities are as follow:

Output 1.1: 450 pumps installed as part of a demonstration phase with KOICA fund. Output 1.1 activities will be accomplished within 30 months of the project life time.
Activities 1.1.1: Selection of 450 farmers to receive the pilot pumps
Activities 1.1.2: Installation of baseline monitoring equipment to establish baseline diesel consumption, water pumped, operating hours, cost expenditure and socio-economic condition
Activities 1.1.3: Specification, procurement, and installation of 450 pilot pumps
Activities 1.1.4: Monitoring performance of the pilot pumps
The Ministry of Water Resources Irrigation and Electricity will coordinate with the Ministry of Finance and will set up the River Nile State PV fund and link with National PV fund which is planned to establish in UNDP/GEF funded project. This activity will be carried out in 2nd year of the project life time and the outputs and detailed activities are as follows:

Output 1.2: Prepared a pilot financial scheme that the River Nile State PV fund can refer to and the seed money on the condition that the financial arrangement of the fund is consistent with the pilot financial model
Activities 1.2.1: Support to formulation on pilot financial scheme for River Nile State PV fund with appropriate fiduciary and legal standards in place for operation and monitoring
Activities 1.2.2: Coordination with River Nile Government State and commercial banks to enact a loan program for PV pumps linked to the PV fund
Activities 1.2.3: Establish a set of criteria for PV pump loans
Activities 1.2.4: Establish and maintain a monitoring system

The Project will coordinate between State Ministry of Agriculture and Agriculture Research Centers (i.e. Head Quarter level and Al-Bashir Wad Medani Agriculture Training Center) and provide technical assistance to the farmers for efficient and effective use on water resources at the farmland under the outcome 2. Agriculture Research Center will support technical assistance for structure development (i.e. small-scale irrigation technique, soil conservation, management and maintenance of the irrigation structure and training about cropping pattern) for the water efficiency and agronomic practices based on the locally adapted technologies. ARCs will train to the extension staff from ministry of agriculture and farmer leaders. The replication of technical know-how scheme will conduct through ToT approach to other farmers. The activities will introduce in the 1st year of project life time and will continue support to the farmers as needed. The outputs and the detailed activities are as follow:

Output 2.1: Enhanced farmers and agriculture extension staff member's skills and knowledge in soil, crop, agronomic practice and water management to undertake sustainable farm management
Activities 2.1.1 Training for specialized engineers and extension staff from ministry of agriculture for small scaled irrigation techniques, soil conservation, management and maintenance of the irrigation infrastructure and solar pumping stations
Activities 2.1.2: Technical studies on shallow groundwater-based irrigation schemes for small farming plots

Activities 2.1.3: Training for climate adaptive agronomics practices to 450 farmers (i.e. selection of crops; cropping pattern; book keeping; harvesting and post harvesting techniques; storage techniques; agroforestry techniques; horticulture techniques)

Output 2.2: Improved efficiency of water usage by using different types of small scale irrigation methods, management and maintenance of the irrigation infrastructure and solar pumping stations delivered to benefit 450 farmers
Activities 2.2.1: Established 45 demonstration plots for efficient use of water techniques
Activities 2.2.2: Provide equipment for small scaled irrigation techniques, soil conservation, management and maintenance of the irrigation infrastructure and solar pumping stations for demonstration plots
Activities 2.2.3: Organise farm demonstrations to showcase the results of the project intervention
Activities 2.2.4: Established additional 405 demonstration plots for efficient use of water techniques

The Project will coordinate between Agriculture Research Center, University of Khartoum and Ahfad Women University and collect ground data, analyze and prepared technical report and advocacy products for replication of the project activities. The activities will carry out after implementing the demonstration plots starting from 2nd year of the project period. The outputs and the detailed activities are as follow:

Output 3.1: Successfully demonstrated practices (type of solar water pumps and adaptation measures in farmland) codified and documented
Activities 3.1.1: Development of small scale irrigation methods, soil conservation, management and maintenance of the irrigation infrastructure with the technical support from Agricultural Research Centers and agriculture training school
Activities 3.1.2: Maintenance of the solar pumping stations
Activities 3.1.3: Dissemination of such measures to farmers to reduce their need for water and thus capacity and capital cost of a pump
Activities 3.1.4: Guidelines on best practices on decentralized water management in farmland

Output 3.2: Designed structured replication programme for other states (at least two states) and integrated into the state level/or agriculture sector development plan
Activities 3.2.1: Farmer to farmer exchange programme from additional targeted states
Activities 3.2.2: Field visit for high level officials and technical specialists from additional targeted states and Khartoum Level
Activities 3.2.3: A comprehensive report documenting lessons learned and successfully tested solar pumping and water management climate adaptive measures in farmland
Activities 3.2.4: Preparation of a package of solar pumping and water management techniques for the integration into the agriculture and food security related programmes and policies, including national and state level programme budgets for further up-scaling
Activities 3.2.5: Lessons codified and disseminated through the Adaptation Learning Mechanism, KOICA and other knowledge networks

III. RESULTS AND PARTNERSHIPS

Expected Results: The project will set up 450 solar pumps in River Nile State in the demonstration stage and established State PV fund. Detailed structure, estimated costs are attached in annex. The design and solar powered pumping system sizing is based on the crop water requirements measured in m^3/day which depend on the size of farms, type of crop and season of cultivation. This information together with the total pumping head (depth of water, draw and losses) will be used to determine the exact required module powers and number of modules that fit the power requirement for the appropriate design of solar pump for each selected farm.

The project will consider past history of floods and other natural disasters which will be included in the Invitation To Bid (ITB) for contractors. When advertising the for the procurement of the solar pumps, the required Hight of panel from the ground will be requested in the design. In addition, the data logger will specifically be designed to operate in remote signal locations and to record the

'vital signs' of the solar pumps. Data is recorded on hourly which is accessible by downloading it in lab top through an installed software program anytime. A pilot tests and calibration of sensors will be conducted with the logger in the field to ensure right data are recorded and accessible.

Gender: In Sudan, women carry out a major portion of agricultural activities and bear almost the entire burden of household work, including water and fuelwood collection and food processing and preparation. In River Nile State, women are active in agriculture within and outside their households. Although women have equal access to land use, very few have land ownership rights, and thus can't access credit, membership in cooperatives or extension. At the household level, women are responsible for a wide range of decision making in farming activities, even when the husband is present (UNEP, 1994).

The project will support to improve participation of women in agriculture by removing one of the barriers which is the labor-intensive operation of diesel pumps. Women account for 51% of the rural population, and agriculture accounts for 78% of the jobs held by rural working women. In generating positive socio-economic impacts for the smallholder agriculture sector, the project will serve to create jobs and raise incomes for women. Women will also benefit, alongside other stakeholder groups, from capacity development activities supported by the project specially under outcome 2 of output 2.1.

The first activities under this outcome are the identification of at least 450 farmers to host 450 solar PV pumps on a "pay if satisfied" basis, and the appropriate specification of these pumps. In selecting the targeted farmers, the project will solicit the applications from the farmers, implement the baseline survey on those applicants, analyze the collected data, set the selection criteria in addition to the geographical distribution and then select the farmers. The pumps will be installed at carefully selected farms[3] at no initial cost to the farmers and women are equally to apply. After one year of operation, if the farmers are satisfied, they can enter into a loan re-payment scheme and benefit from the subsidy offered. The re-payment funding will be used as the seed funding for the national PV Fund, once established. If the farmers are not satisfied, the pumps will be removed at no cost to them. In effect, the famers who receive the 450 pumps benefit from a free one-year trial, and if they continue, a one-year grace period on their loan. This method achieves several objectives. It allows the farmers to try the pumps with no risk to them and popularize the technologies and provide a proof-of-concept. At the same time, they realize that if they choose to keep the pump they will have to pay for it; thus they have an incentive to honestly evaluate its performance and to ensure its safety. The project principal is "not left behind" and will take into account small holder farmers (men and women) from diverse agro-ecological zones of the River Nile State including desert and semi-desert zones.

The objectives of the 450 pilot pumps are: a) to make farmers familiar with solar technology and enable them to visit operating pump installations, see them in operation, and obtain feedback from fellow farmers; b) to provide data on the operation of the pumps to the project team which will inform the design and selection of future pumps; and c) provide confidence to other stakeholders, such as lenders, on the effectiveness of solar pumps.

Upon selection of a farmer to host one of the trial pumps (to be done by the River Nile State PV pump committee and Farmer's Association based on transparent criteria), the farmer will enter into a contract with project stipulating that the farmer will have the choice at the end of the year to either have the pump removed at no cost or purchase it. If the farmer decided to purchase the pump, the payment will return to the National PV fund. Farmer selection process will base on the agreed criteria and the selection process will conduct transparent manner by River Nile State PV fund (including Farmer's Association) approved by the project. The farmers will be meet certain criteria specified in advance, such as: they fit a geographical distribution to allow the greatest number of other farmers to see pumps in operation; they are a respected person within the community such that their testimony to the efficacy of PV pumps holds weight; the banks are prepared to enter into an agreement with them for finance of the solar PV pumps; and they agree

[3] Farms to be selected to be representative and owned by individuals known to be trustworthy and upstanding members of the community.

to allow visitors to observe the pump and to collect all data from their present pumping activity and future PV pump available to the project and publicly (with appropriate anonymity).

The project can then collect data on the operation of the diesel pumps in the time period between when the farmers are identified and when the solar PV pumps are installed. Once farmers are identified, diesel and water flow meters will be installed to establish a more specific baseline. In addition, to evaluate the impact of the solar pumps, the project will conduct more comprehensive baseline measurements such as not only for the water and fuel costs but also the farming practices, the area of cultivated land, crops and flow of income from agriculture. Farmers for the pilot phase are expected to be identified as soon as is practical after the project start. As it will take a few months for the solar PV pumps to be selected, procured, imported and installed, this should provide at least 3-6 months of data on the operation of diesel pumps to be replaced by a solar pump.

In order to be eligible to receive finance from the project at demonstration stage, farmers should meet at least the following minimum criteria which is based on the past experiences of the projects but not limited to:

a. Have a farm in an area that the electricity grid will not reach within the coming 10 years
b. Sign a declaration allowing the Project access to data from the pump for development purposes
c. Sign a declaration that the farmer will not sell or otherwise use the PV system and pump for anything other than the intended purpose.
d. Select a PV system which meets the criteria established (such systems will be identified by NERC)
e. Participation in a water efficiency programme
f. Meet any credit-worthiness or other criteria imposed by the project

As a participant in establishment of the River Nile PV Fund, the UNDP-KOICA project will have the opportunity to participate in creating equitable criteria and ensuring they are adequately applied.

Partnerships and Stakeholder Engagement:

The main stakeholders in the project are as follows:

- Ministry of Water Resources Irrigation and Electricity (MWRIE)
- National Energy Research Center (NERC)
- State Ministry of Economic Resources and Production - River Nile State
- Ministry of Finance & National Economy (MoF)
- Ministry of Agriculture (MoA)
- Higher Council for Environment and Natural Resources (HCENR)
- KOICA and
- The Banks

The **Ministry of Water Resources Irrigation and Electricity (MWRIE)** is the Government body responsible for electric power in Sudan. MWRE has been implementing the grid-electric pumps programme together with the Northern Sates government to promote electric pumping for large-scale farms. MWRIE will be the main project's stakeholder for the project. MWRIE is the responsible for the provision of technical specification for the installed solar pumps and together with the state authorities, will assist in the site selection for the participating farmers.

The **State Ministry of Economic Resources and Production - River Nile State** is the governmental entity which is responsible for the implementation of the Agricultural Strategic Plan in the River Nile State. The main target of the plan is to raise the percentage of agricultural land in the country by 70%. The River Nile State MoA acts as the body responsible for the overall management of agricultural affairs in the River Nile State and is the primary State Government liaison with the Farmer's Union of the River Nile State.

Higher Council for Environment and Natural Resources (HCENR) - The Higher Council for Environment and Natural Resources oversees the application of environmental laws and regulations to all development projects in Sudan, and has particular responsibilities in the climate change area. HCENR serves as the Designated National Authority (DNA) for the Clean Development Mechanism (CDM). It is also the NAMA Focal Point and UNFCCC Focal Point for Sudan. With UNDP support, HCENR has been developing standardized baselines for Sudan. HCENR has also developed a Technology Needs Assessment (TNA) for Climate Change Adaption and Mitigation, funded by the GEF. The Secretary General of HCENR serves as the National Designated Authority of the Green Climate Fund.

The **Ministry of Finance & National Economy (MoF)** has the general objective of developing the internal resources of Sudan and utilizing them in the most efficient way possible to support growth. The MoF also directs the customs and tax authorities and thus is responsible for taxation and for exempting strategic goods from customs duties and taxes, as agricultural implements are, for example.

The **National Energy Research Centre (NERC)** has been active in promoting and developing solar water pumping. NERC has a special department for solar energy equipped with instruments and a mechanical workshop. NERC has already participated in the installation of solar pumps around Sudan and is one of the most experienced entities in this regard in the country. NERC is tasked with the development of Sudan's future energy resources and securing the energy needed for sustainable growth. It is hosted within the Ministry of Science and Communication.

The **Farmer's Union of the River Nile State** serves as the body representing Farmers' interests within the state and interfacing with the State Government. The Union coordinates with the River Nile State Ministry of Agriculture on matters relating to farmers' demands and implementation of national programmes which impact farmers in the River Nile State. The head of the Farmer's Union also represents the Union at workshops and stakeholder consultations giving a unified voice to farmers.

The KOICA will play key role in i) part of the Project Board members, ii) participate in technical, inception and decision-making workshops and meetings, iii) monitoring field visits, and iv) review the progress reports and gave comments for future direction.
The banks provide general loans and finance to clients and in the case of solar water pumping will combine to form a fund (mahfaza) to support solar water pumping under unified terms. The banks in the River Nile State expected to participate are: Bank of Sudan, Agricultural bank, Central Islamic bank, Agricultural & commercial bank, Al Nile Bank, Sudanese Islamic Bank, Baraka Bank, Al Shamal Islamic Bank, Farmer's Commercial Bank and Family Bank.

Sustainability: The sustainable adoption of PV pumping will depend largely on two factors. The first is the technical ability of the pumps to deliver pumping to the satisfaction of the farmers. There may be some learning with regards to this aspect, but it is expected that this is a readily solvable technical problem, in particular if the farmers as willing to adopt water saving measures. The second factor is the availability of adequate financing. This is in most circumstances and locations around the world a determining factor. Thus, the Project will provide specific consideration for the PV financing mechanism with the participation of commercial banks and allocate the bulk of project resources towards developing a sustainable financial mechanism and providing subsidies to encourage adoption of the solar pumps. The subsidy levels needed will be determined dynamically and evolve throughout the project.

An important factor in the sustainability of solar PV pumping is the cost of diesel, as it is the main alternative. Diesel remains subsidized in Sudan though subsidies have been reduced in the past and are expected to be lifted completely in the future though the time schedule for this is not clear. If such subsidies are lifted solar pumping would become considerably more economical than diesel and with an appropriate financing mechanism in place would be expected to see widespread adoption.

The project will provide mechanisms to reduce the up-front risks of adoption of PV pumps for all market participants: farmers, financiers, and suppliers. It also has the benefit of reducing diesel

consumption therefore reducing the burden of supply on the national infrastructure and reducing the subsidy burden on the State budget (for as long as diesel is subsidized).

The Project will take a market-based approach supporting a regulated supply-chain, building technical expertise, and a financing mechanism. With these elements in place, the adoption of PV pumping can be expected to be self-sustaining. The major risk to this sustainability is the stability of the Sudanese pound. If the pound loses value, the price of solar pumps increases as they are imported. If diesel continues to be subsidized, the subsidy burden on the State increases but the attractiveness of solar pumping to individual farmers is reduced.

In addition to the sustainability from the energy perspective the water saving measures that the project will put in place will have the effect of reducing water withdrawal from the underground aquifer thereby increasing its longevity.

Finally, the project will focus on a structured replication programme and knowledge sharing programme to replicate success in the River Nile to other States of Sudan. The project fund will use for structured replication programme for other states and codified, documented and share best practices (type of solar water pumps and adaptation measures) to and integrated into the state level and national policies. The concessions will have national force and will, therefore, also promote nationwide take-up of PV pumps with the support of KOICA fund. Replication will be encouraged in other states and areas by demonstrating, documenting, and disseminating the avoided costs of infrastructure and fossil fuels, as well as the environmental benefits of avoiding diesel use, thereby incentivizing governments in other states to adopt measures, as outline in output 3.2.

Risks and Assumptions: In accordance with KOICA's SO1 "Promote Sustainable Development through Climate Change Response", the key outcomes and success indicators of the project are:

- Reduction in Greenhouse Gases (Outcome 9.1.1); Indicator: the amount of CO_2 reduced (ton CO_2 per eq)
- Improvement in climate adaptive capacity (Outcome 9.1.2); Indicator: the distribution rate of service water and irrigation water % and the increase rate of major crops production %

The project specifically aims to achieve the following:

- Installation of 450 pilot solar pumps funded by the KOICA Project to be re-paid only if the farmer/owner is satisfied with performance
- Creation of a River Nile PV Fund as a financing mechanism for solar PV pumps
- Development of local capacity to size, install, and service solar pumps
- Implementation of water efficient irrigation techniques to promote the long-term sustainability of water use
- Replication of project components executed in the River Nile State in Sudan's other states

The main **risks** identified to the successful implementation of the project include:

Currency risk: Diesel price is fixed in local currency. Pumps are imported without price fixing and thus their price fluctuates with the local currency exchange rate. If the value of local currency falls, solar PV pumps will become increasingly unattractive compared with diesel pumps. This risk is mitigated by the Government's intentions to remove all subsidies from diesel. However, this has not yet occurred and no clear timeline for implementation has been announced. The project will seek to mitigate this risk by removing taxes and import duties on solar PV pumps, and providing low-cost financing therefore making them more competitive and allowing a larger margin for currency fluctuation before solar PV pumps are unattractive compared with diesel.

Falling oil prices: which is in the same manner as currency risk, impact the project. Reduced international oil prices make the continuation of diesel subsidies in Sudan more likely as the cost of subsidy to Government is greatly reduced, or even removed. Long-term continuation of subsidies would threaten the long-term sustainability of the project. At present a small subsidy for PV pumps, provided by the KOICA project, makes solar PV pumps financially attractive. If diesel subsidies are not lifted by the time the end of the project when the KOICA subsidy stops, pumps

may be financially unattractive compared with diesel. If PV pumps are adopted as a viable technology, and efficient financing mechanisms exist, farmers may be willing to adopt PV pumps even at a premium to diesel pumps in order to avoid the difficulties associated with operating a diesel pump.

Government support: The project relies on Government support for the establishment of the River Nile State PV fund, as well as for modifications to legislation to include solar PV pumps among tax and customs duties exempt equipment. If such Government support should fail to materialize the project would run the risk of pumps being unattractive compared with the alternative. This risk is considered overall as low. A co-finance letter committing Ministry of Finance support will obtain in full-fledged project document formulation period. The adoption of solar PV pumps reduces diesel consumption therefore reducing the overall diesel subsidy burden on the Government. So, the Government has good incentives to support the project.

Climate Change Risk: Climate change risk is manifest mainly in the risk of reduced water source for pumping, or reduced rainfall requiring increased pumping. Sudan presently does not utilize its entire allocation of Nile river water. Both scenarios, whether reduced rainfall, reduced water levels, or increased temperature require additional pumping capacity to support crops. Solar PV pumps are to a large extent modular. Thus, increased pumping capacity, to an extent, may be realized with the addition of solar PV modules to existing pumping systems.

Technology: Technical risk is minimal. The main technical risk is in ensuring quality components fit for the purpose. The main counterpart in this activity will be NERC who together will determine standards for pumps to be financed by the banks and develop a list of “approved” components.

Performance risk: Solar pumping reduces irrigation hours to those when the sun is sufficiently strong. Compared with the use of diesel which allows pumping independent of the weather, solar pumping may not deliver the volume of water farmers are used to, or the pumps may need to be over-sized compared with the present diesel pumps. A combination of capacity building and testing to appropriately size the pumps and adoption of efficient irrigation techniques will help overcome this risk.

Implementation capacity: Inadequate and/or non-capacitated human resources to successfully implement the project and support the mainstreaming of its results. The current capacity to implement and operate solar pumping in Sudan is very low because the market is small. However, with training (to be provided by the project) those with basic technical skills can learn the skills necessary to install and service solar pumps.

Project management risks: a committed, full-time National Project Manager with adequate outreach and networking skills is absolutely essential for the success of the activities. The National Project Manager should have an ability: i) to engage the key stakeholders in constructive discussion about future renewable energy development needs; ii) to guide and supervise the studies undertaken and effectively co-operate with the international experts who are engaged to support this work; iii) to present their findings and recommendations in a convincing manner to key policy-makers and opinion leaders by taking into account the main macroeconomic and policy drivers for development of the solar pumping sector development; and iv) to identify areas of future work. During project implementation, the National Project Manager also needs to be supported by qualified technical and legal committee.

IV. PROJECT MANAGEMENT

The project will be directly executed by UNDP through Direct Implementation Modality (DIM). UNDP will be accountable for the disbursement of funds and the achievement of the project goals, according to the approved work plan in consultation with Ministry of Water Resources Irrigation and Electricity. UNDP will also be using Letters of Agreement (LOA) with (Ministry or Water Resources, Irrigation, and Electricity (MWRIE) to support the implementation of output 1, State Ministry of Economic Resources and Production- River Nile State, to support the implementation of output 2. A Government Project Coordinator (GPC) will be appointed by the above government entities to manage the LOAs, coordinate project operations and support the Project Manager (PM)

with the overall administration, oversight, coordination of activities and maintaining liaison with UNDP and the other Government entities. Their remuneration will be incurred by the government. The Ministry of Finance and Economic Planning will engage with UNDP in the regular monitoring of progress of field activities, expenditures and achievement of results. The Ministry of Finance and Economic Planning will also monitor the reimbursement of solar pumps cost and the functionality of the PV fund and will assist is solving any issues related to the National PV Fund.

V. RESULTS FRAMEWORK[4]

Intended Outcome as stated in the UNDAF/Country Programme Results and Resource Framework:

UNDAF/CPD outcome 2: By 2021, people's resilience to consequences of climate change, environmental stresses and natural hazards is enhanced through strengthened institutions, policies, plans and programmes

Outcome indicators as stated in the Country Programme Results and Resources Framework, including baseline and targets:

UNDAF/CPD Outcome indicator 2.1 (Resilience indicator)
Output indicator 3.1. Output from smallholder agriculture
Output 3.1. Access to clean energy for the poor enhanced.
Indicator: 3 1.1: Number of households supported to access clean energy (IRRF 1.5.2)

Applicable Output(s) from the UNDP Strategic Plan:

IRRF Output 1.5: Inclusive and sustainable solutions adapted to achieve increased energy efficiency and/or sustainable energy solutions targeting undiscovered communities/groups and women.

Indicator 1.5.1: Number of new development partnerships with funding for improved energy efficiency and/or sustainable energy solutions targeting underserved communities/groups and women.

Indicator 1.5.2: Extent of change in: a) energy efficiency and/or b) modern energy coverage by users and specific sectors

Project title and Atlas Project Number: Award ID – 00106169; Output ID – 00107041

Solar water pumps for Sustainable Agriculture in Sudan: Promotion of renewable energy and sustainable financial modality as gateway to improve farmers' livelihoods and GHG reduction

EXPECTED OUTPUTS	OUTPUT INDICATORS[5]	DATA SOURCE	BASELINE		TARGETS (by frequency of data collection)						DATA COLLECTION METHODS & RISKS
			Value	Year	Year 1	Year 2	Year 3	Year 4	Year …	FINAL	

[4] UNDP publishes its project information (indicators, baselines, targets and results) to meet the International Aid Transparency Initiative (IATI) standards. Make sure that indicators are S.M.A.R.T. (Specific, Measurable, Attainable, Relevant and Time-bound), provide accurate baselines and targets underpinned by reliable evidence and data, and avoid acronyms so that external audience clearly understand the results of the project.

[5] It is recommended that projects use output indicators from the Strategic Plan IRRF, as relevant, in addition to project-specific results indicators. Indicators should be disaggregated by sex or for other targeted groups where relevant.

Output 1 Financing and dissemination mechanism established and operational to support a PV pump installation programme	***1.1*** Investment mobilized for purchase of solar pumps	Terminal impact assessment	0	US$ 3,718, 013	X						Commercial banks are willing to co-finance based on potential of solar water pump replication and MF regulations; State Government in collaboration with Ministry of Finance including Banks and set up State PV fund
	1.2 Dedicated mechanism for finance of PV pumps established	Interviews with banks, farmers, and suppliers.	1	At least one State PV pump fund		X					
Output 2 Improved in efficiency of water usage in the farm land	2.1 Number of pumping system using water efficient irrigation methods	Report on water consumption and pumped volumes	28	450	X	X	X	X			The use of water at present is not optimal and substantial improvements can be made. Farmers are willing to undertake water efficiency techniques in current farming practices; Local governments are supportive and engaged in implementation process; Extension workers and farmers for integrating and addressing climate risks into livelihood activities are willing to participate in the training workshops and recognize the benefits in engaging in adaptation processes to ensure crop yield improvement and food security; Extreme drought events do not disrupt the implementation of the adaptation measures
	2.2 Number hectare improved in water efficiency	Project progress and State Ministry of Agriculture reports	0	At least 1125 hectar es	X	X	X	X			

Output 3 Supportive enabling environment and scaled-up implementation	**3.1** PV Pumping integrated in State Development plan or sectoral plan	Project Terminal evaluation report	0	2			X				Cooperation of Government, State Agriculture Ministry and regulatory bodies
	3.2 Documented best practices and disseminated to development partners	Project Terminal Evaluation Report	0	5				X			

VI. MONITORING AND EVALUATION

In accordance with UNDP's programming policies and procedures, the project will be monitored through the following monitoring and evaluation plans: *[Note: monitoring and evaluation plans should be adapted to project context, as needed]*

Monitoring Plan

Monitoring Activity	Purpose	Frequency	Expected Action	Partners (if joint)	Cost (if any)
Track results progress	Progress data against the results indicators in the RRF will be collected and analysed to assess the progress of the project in achieving the agreed outputs.	Quarterly, or in the frequency required for each indicator.	Slower than expected progress will be addressed by project management.		8,400
Monitor and Manage Risk	Identify specific risks that may threaten achievement of intended results. Identify and monitor risk management actions using a risk log. This includes monitoring measures and plans that may have been required as per UNDP's Social and Environmental Standards. Audits will be conducted in accordance with UNDP's audit policy to manage financial risk.	Quarterly	Risks are identified by project management and actions are taken to manage risk. The risk log is actively maintained to keep track of identified risks and actions taken.		10,000
Learn	Knowledge, good practices and lessons will be captured regularly, as well as actively sourced from other projects and partners and integrated back into the project.	At least annually	Relevant lessons are captured by the project team and used to inform management decisions.		
Annual Project Quality Assurance	The quality of the project will be assessed against UNDP's quality standards to identify project strengths and weaknesses and to inform management decision making to improve the project.	Annually	Areas of strength and weakness will be reviewed by project management and used to inform decisions to improve project performance.		
Review and Make Course Corrections	Internal review of data and evidence from all monitoring actions to inform decision making.	At least annually	Performance data, risks, lessons and quality will be discussed by the project board and used to make course corrections.		28,000
Project Report	A progress report will be presented to the	Annually, and at			

	Project Board and key stakeholders, consisting of progress data showing the results achieved against pre-defined annual targets at the output level, the annual project quality rating summary, an updated risk long with mitigation measures, and any evaluation or review reports prepared over the period.	the end of the project (final report)			28,000
Project Review (Project Board)	The project's governance mechanism (i.e., project board) will hold regular project reviews to assess the performance of the project and review the Multi-Year Work Plan to ensure realistic budgeting over the life of the project. In the project's final year, the Project Board shall hold an end-of project review to capture lessons learned and discuss opportunities for scaling up and to socialize project results and lessons learned with relevant audiences.	Specify frequency (at least annually)	Any quality concerns or slower than expected progress should be discussed by the project board and management actions agreed to address the issues identified.		28,000

Evaluation Plan[6]

Evaluation Title	Partners (if joint)	Related Strategic Plan Output	UNDAF/CPD Outcome	Planned Completion Date	Key Evaluation Stakeholders	Cost and Source of Funding
Mid-Term Evaluation	Nil	SP output # 1.5	CPD outcome #2	March 2022	MWRIE, Banks and small land holder farmers	USD 35,000 KOICA fund
Terminal Evaluation	Nil	SP output # 1.5	CPD outcome #2	Jan 2023	MWRIE, Banks and small land holder farmers	USD 35,000 KOICA fund

[6] Optional, if needed

VII. MULTI-YEAR WORK PLAN [7][8]

All anticipated programmatic and operational costs to support the project, including development effectiveness and implementation support arrangements, need to be identified, estimated and fully costed in the project budget under the relevant output(s). This includes activities that directly support the project, such as communication, human resources, procurement, finance, audit, policy advisory, quality assurance, reporting, management, etc. All services which are directly related to the project need to be disclosed transparently in the project document.

EXPECTED OUTPUTS	PLANNED ACTIVITIES	Planned Budget by Year				RESPONSIBLE PARTY	PLANNED BUDGET		
		Y1	Y2	Y3	Y4		Funding Source	Budget Description	Amount
Output 1: Financing and dissemination mechanism established and operational to support a PV pump installation programme Output 1.1. 450 pumps installed as part of a demonstration phase Output 1.2. Prepared a pilot financial scheme that the River Nile State PV fund can refer to and the seed money on the condition that the financial arrangement of the fund is consistent with the pilot financial model *Gender marker:*	1.1 .1 Selection of 450 farmers to receive the pilot pumps	X				UNDP	KOICA	Training, workshops and conference	15,000
	1.1.2 Installation of baseline monitoring equipment to establish baseline diesel consumption, water pumped, operating hours, cost expenditure and socio-economic condition	X				UNDP	KOICA	Materials and goods	72,000
	1.1.3 Specification, procurement, and installation of 368 pilot pumps	X	X	X		UNDP	KOICA	Grants	2,649,707
	1.1.3 Specification, procurement, and installation of 82 pilot pumps	X	X	X		UNDP	UNDP	Grants	600,000
	1.1.3 Technical Specialist contribute 100% of time to output 1, 2 and 3					UNDP	KOICA		429,492
	1.1.4 Conduct monitoring performance of the pilot pumps	X	X	X		UNDP	KOICA	Contractual Services – Companies / Nat-Serv	45,000
	1.2.1 Support to Ministry of Finance to create River Nile State PV fund with appropriate fiduciary and legal standards in place for operation and monitoring		X			UNDP	KOICA	International consultants	87,750

[7] Cost definitions and classifications for programme and development effectiveness costs to be charged to the project are defined in the Executive Board decision DP/2010/32

[8] Changes to a project budget affecting the scope (outputs), completion date, or total estimated project costs require a formal budget revision that must be signed by the project board. In other cases, the UNDP programme manager alone may sign the revision provided the other signatories have no objection. This procedure may be applied for example when the purpose of the revision is only to re-phase activities among years.

	1.2.2 Coordination with River Nile Government State and commercial banks to enact a loan program for PV pumps linked to the PV fund		X			UNDP	KOICA	Local consultants	20,250
	1.2.3 Establish a set of criteria for PV pump loans		X			UNDP	KOICA	International consultants	87,750
	1.2.4 Establish and maintain a monitoring system		X			UNDP	KOICA	Local consultants	20,250
	1.2.4. Programme Analyst (Energy & Climate Change) contribute 30% of time for output 1, 2 and 3 for project life	X	X	X	X	UNDP	KOICA	Programme staff salary	192,172
	Sub-Total for Output 1								4,219,371
Output 2: Improved in efficiency of water usage in the farm land Output 2.1. Enhanced farmers and agriculture extension staff member's skills and knowledge in soil, crop, agronomic practice and water management to undertake sustainable farm management Output 2.2. Improved efficiency of water usage by using different types of small-scale irrigation methods, management and maintenance of the irrigation infrastructure and solar pumping stations delivered to benefit 450 farmers *Gender marker:*	2.1.1 Training for specialized engineers and extension staff from ministry of agriculture for small scaled irrigation techniques, soil conservation, management and maintenance of the irrigation infrastructure and solar pumping stations	X				UNDP	KOICA	Contractual Services - Companies / Nat-Serv	7,500
	2.1.2 Technical studies on shallow groundwater-based irrigation schemes for small farming plots	X				UNDP	KOICA	Materials & Goods	29,250
	2.1.3 Training for climate adaptive agronomics practices to 450 farmers (i.e. selection of crops; cropping pattern; book keeping; harvesting and post harvesting techniques; storage techniques; agroforestry techniques; horticulture techniques)	X	X	X		UNDP	KOICA	Contractual Services - Companies / Nat-Serv	67,500
	2.2.1 Established 45 demonstration plots for efficient use of water techniques	X				UNDP	KOICA	International Consultants	18,000
	2.2.2 Provide equipment for small scaled irrigation techniques, soil conservation, management and maintenance of the irrigation infrastructure and solar pumping stations for demonstration plots	X	X	X		UNDP	KOICA	Contractual Services - Companies / Nat-Serv	180,000
	2.2.3 Organise farm demonstrations to showcase the results of the project intervention		X	X	X	UNDP	KOICA	Contractual Services - Companies / Nat-Serv	45,000

	2.2.4 Established additional 405 demonstration plots for efficient use of water techniques		X	X	X	UNDP	KOICA	Contractual Services - Companies / Nat-Serv	162,000
	2.2.4 Programme Specialist (Sustainable Livelihoods) contribute 30% of time for output 1, 2 and 3 for project life	X	X	X	X	UNDP	KOICA	Programme staff salary	158,095
	Sub-Total for Output 2								667,345
Output 3: Supportive enabling environment and scaled-up implementation Output 3.1. Successfully demonstrated practices (type of solar water pumps and adaptation measures in farmland) codified and documented Output 3.2. Designed structured replication programme for other states (at least two states) and integrated into the state level/or agriculture sector development plan *Gender marker:*	3.1.1 Development of small-scale irrigation methods, soil conservation, management and maintenance of the irrigation infrastructure at the Agricultural Research Centre in the River Nile State	X	X			KOICA	KOICA	International Consultants	237,250
	3.1.2 Maintenance of the solar pumping stations	X	X	X	X	UNDP	KOICA	Contractual Services - Companies / Nat-Serv	45,000
	3.1.3 Dissemination of such measures to farmers to reduce their need for water and thus capacity and capital cost of a pump	X	X	X	X	UNDP	KOICA	Contractual Services - Companies / Nat-Serv	45,000
	3.1.4 Guidelines on best practices on decentralized water management in farmland	X	X			KOICA	KOICA	International Consultants	42,250
	3.1.4 M&E (Mid and Terminal)		X		X	UNDP	KOICA	International Consultants	70,000
	3.2.1 Farmer to farmer exchange programme from additional targeted states	X	X	X	X	UNDP	KOICA	Training, Workshops and Conference	6,750
	3.2.2 Korea visits of high-level officials to learn lessons		X	X	X	UNDP	KOICA	Training, Workshops and Conference	7,000
	3.2.2 Korea visits of high-level officials to learn lessons		X			KOICA	KOICA	Training, Workshops and Conference	22,500
	3.2.3 A comprehensive report documenting lessons learned and successfully tested solar pumping and water management climate adaptive measures in farmland				X	UNDP	KOICA	International Consultants	42,250

	3.2.4 Preparation of a package of solar pumping and water management techniques for the integration into the agriculture and food security related programmes and policies, including national and state level programme budgets for further up-scaling			X	X	KOICA	KOICA	International Consultants	87,750
	3.2.5 Lessons codified and disseminated through the Adaptation Learning Mechanism, KOICA and other knowledge networks				X	KOICA	KOICA	International Consultants	40,250
	3.2.5 Monitoring, Korea Experts and training costs managed by KOICA	X	X	X	X	KOICA	KOICA		400,000
	3.2.5 Programme Specialist (Energy, Environment and Climate change) contribute 30% of time for output 1, 2 and 3 for project life	X	X	X	X	UNDP	KOICA	Programme staff salary	249,733
	Sub-Total for Output 3								1,295,733
General Management Support	4.1.2 National Project Manager (SB4/5)	X	X	X	X	UNDP	KOICA	Contractual Services - Individ	48,000
	4.1.3 Finance and Admin Associate (SB2/3)	X	X	X	X	UNDP	KOICA	Contractual Services - Individ	38,400
	4.1.4 Driver, Khartoum (SB2/1)	X	X	X	X	UNDP	KOICA	Contractual Services - Individ	33,600
	4.1.5 M&E and State Engineer	X	X	X	X	UNDP	KOICA	Contractual Services - Individ	84,000
	4.1.6 1 cars (Hilux Double Cab)	X				UNDP	KOICA	Contractual Services - Individ	30,000
	4.1.7 Fuel (vehicle and generator)	X	X	X	X	UNDP	KOICA	Materials & Goods	4,800
	4.1.8 License renewal and insurance (1 car)	X	X	X	X	UNDP	KOICA	Materials & Goods	1,000
	4.1.9 Repair and maintenance (car)	X	X	X	X	UNDP	KOICA	Materials & Goods	1,920
	4.1.10 Duty Travel Actual Travel expenses (KRT 2 programme staff, KRT 2 project staff and 1 State staff)	X	X	X	X	UNDP	KOICA	Supplies	18,400
	4.1.11 Stationary	X	X	X	X	UNDP	KOICA	Travel	4,800

	4.1.12 Toner/cartridge	X	X	X	X	UNDP	KOICA	Supplies	1,800
	4.1.13 Generator (servicing & repair)					UNDP	KOICA	Supplies	2,500
	4.1.14 Copier and Printer maintenance	X	X	X	X	UNDP	KOICA	Supplies	1,000
	4.1.15 Security equipment	X				UNDP	KOICA	Supplies	4,300
	4.1.16 Internet usage & VSAT	X	X	X	X	UNDP	KOICA	Materials & Goods	3,655
	4.1.17 Electricity bills and water Bills	X	X	X	X	UNDP	KOICA	Information Technology Equipment	4,800
	4.1.18 Phone call bills	X	X	X	X	UNDP	KOICA	Supplies	5,040
	4.1.19 Office equipment purchase (IT, Printer, copier) and furniture	X				UNDP	KOICA	Communic & Audio Visual Equip	26,250
	4.1.20 Services provided by operations (i.e. Finance, procurement, HR, ICT)	X	X	X	X	UNDP	KOICA	Services to project (GoE)	34,993
	Sub-Total for Output 4								349,258
TOTAL	Total amount of KOICA fund is USD 5,931,707, of this KOICA shall keep for USD 830,000 for M&E visits, technical assistance and training cost in Korea. UNDP shall manage USD only 5570000.								6,531,707
GMS 8%	General Management Services								412,593
Levy 1%	UNDP managed 5,570,000 USD, of these 1% is for Levy Fee								55,700
Grand Total									7,000,000

VIII. GOVERNANCE AND MANAGEMENT ARRANGEMENTS

The project will be directly executed by UNDP through Direct Implementation Modality (DIM). UNDP will be accountable for the disbursement of funds and the achievement of the project goals, according to the approved work plan in consultation with Ministry of Water Resources Irrigation and Electricity. UNDP will also be using Letters of Agreement (LOA) with (Ministry or Water Resources, Irrigation, and Electricity (MWRIE) to support the implementation of output 1, State Ministry of Economic Resources and Production- River Nile State, to support the implementation of output 2. A Government Project Coordinator (GPC) will be appointed by the above government entities to manage the LOAs, coordinate project operations and support the Project Manager (PM) with the overall administration, oversight, coordination of activities and maintaining liaison with UNDP and the other Government entities. Their remuneration will be incurred by the government.

A **Project Board** (PB) will be established at the inception of the project to monitor project progress, guide project implementation and to support the project in achieving its listed outputs and outcomes. The PB remains sufficiently lean to be operationally effective. The final list of the PB members will be completed at the outset of project operations and presented in the Inception Report taking into account the envisaged role of different parties in the PB. The National Project Manager will participate as a non-voting member in the PB meetings and will also be responsible for sharing required documents sufficiently in advance of the meeting and compiling a summary report of the discussions and conclusions of each meeting. The Project Board contains four distinct roles which have been described as follows:

1) **An Executive**: individual representing the project ownership to chair the group. UNDP will chair the board.
2) **Senior Supplier**: group representing the interests of the parties concerned which provide funding for specific cost sharing projects and/or technical expertise to the project. The Senior Supplier's primary function within the Project Board is to provide guidance regarding the technical feasibility of the project and alignment of the outcomes/outputs with the KOICA fund. Ministry of Finance and Economic Planning – Directorate of External Financing – will provide tax exemption for solar systems and ensure contribution for local component. Commercial banks will provide co-financing for National and River Nile State PV fund. The Ministry of Finance and Economic Planning will engage with UNDP in the regular monitoring of progress of field activities, expenditures and will monitor how the PV Fund is functioning.

3) **Senior Beneficiary**: group of individuals representing the interests of those who will ultimately benefit from the project. The Senior Beneficiary's primary function within the Project Board is to ensure the realization of project results from the perspective of project beneficiaries. These are represented by:
    - The Ministry of Water Resources, Irrigation and Electricity Sate Ministry of Agriculture – River Nile State; National Energy Research Centre (NERC), and Agricultural Research Centre (ARC), Other members can be invited at the decision of the PB and as-needed.

Coordination of the above stakeholders will be carried out by UNDP with the support of MWRIE. The PB will identify and put in place steps for initial activities to support, for example, the technical capacity building in the period when the regulatory and financial structures are being developed. One goal of project coordination will be to ensure that the various components of the project are in place when they are needed: e.g. financial instruments are ready when regulations come into place; technical capacity and equipment supply are available at the appropriate time, etc. The PB will meet semi-annually during project implementation, and it will have the responsibility of coordinating and harmonizing the actions of all the key stakeholders.

The day-to-day management of the project will be carried out by a National Project Management Unit (PMU) under the overall guidance of the PB. The PMU will be established within UNDP and will coordinate its work with MWRIE, and other stakeholders. UNDP will recruit a **Technical**

Specialist to provide technical assistance in consultation with other national and international consultants. a **National Project Manager** will be recruited and will be responsible of the day to day management. He/she will report to UNDP and the PB and a Chief Technical Advisor will provide technical assistance to the project team. The project personnel will be selected on a competitive basis in accordance with the relevant rules and procedures of UNDP Country Office.

The Chief Technical Advisor will take a lead role in the implementation of specific technical assistance components of the project in consultation with international and national experts, taking the contacts with experts and institutions in other countries that have already gained experience in developing and implementing renewable energy policies and financial support mechanisms are also to be established.

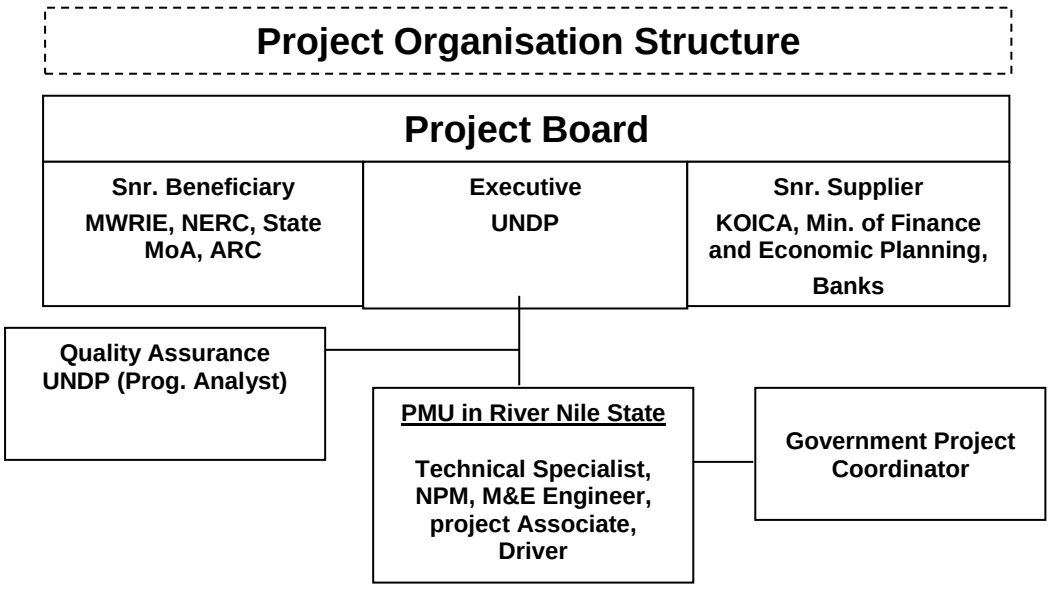
A project Associate will also be recruited to provide the admin and finance support functions.

An Engineer (Monitoring and Evaluation Officer) will be recruited to monitor technical aspects, installation and will also be responsible for the project reporting.

UNDP will maintain the oversight and management of the overall project budget. It will be responsible for monitoring project implementation, timely reporting of the progress to the KOICA, as well as organizing mandatory and possible complementary reviews and evaluations on an as-needed basis. It will also support the executing agency in the procurement of the required expert services and other project inputs and administer the required contracts. Furthermore, it will support the coordination and networking with other related initiatives and institutions in the country.

To successfully reach the objective and outcomes of the project, it is essential that the progress of different project components is closely monitored both by the key local stakeholders and authorities as well as by project's international experts, starting with the finalization of the detailed, component-specific work plans and implementation arrangements and continuing through the project's implementation phase. The purpose of this is to facilitate early identification of possible risks to successful completion of the project together with adaptive management and early corrective action, when needed.

In order to accord proper acknowledgement to the KOICA for providing funding, a KOICA logo should appear on all relevant KOICA project publications, including any hardware purchased with KOICA funds. Any citation on publications regarding projects funded by KOICA should also accord proper acknowledgement to the KOICA in accordance with the relevant KOICA guidelines.



IX. LEGAL CONTEXT

Option a. Where the country has signed the Standard Basic Assistance Agreement (SBAA)

This project document shall be the instrument referred to as such in Article 1 of the Standard Basic Assistance Agreement between the Government of (country) and UNDP, signed on 30/Sept./1981. All references in the SBAA to "Executing Agency" shall be deemed to refer to "Implementing Partner."

This project will be implemented by [name of entity] ("Implementing Partner") in accordance with its financial regulations, rules, practices and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. Where the financial governance of an Implementing Partner does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition, the financial governance of UNDP shall apply.

X. RISK MANAGEMENT

1. UNDP as the Implementing Partner will comply with the policies, procedures and practices of the United Nations Security Management System (UNSMS.)

2. UNDP as the Implementing Partner will undertake all reasonable efforts to ensure that none of the [project funds][9] [UNDP funds received pursuant to the Project Document][10] are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the list maintained by the Security Council Committee established pursuant to resolution 1267 (1999). The list can be accessed via http://www.un.org/sc/committees/1267/aq_sanctions_list.shtml. This provision must be included in all sub-contracts or sub-agreements entered into under this Project Document.

3. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

4. UNDP as the Implementing Partner will: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

5. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

6. UNDP as the Implementing Partner will ensure that the following obligations are binding on each responsible party, subcontractor and sub-recipient:

    a. Consistent with the Article III of the SBAA *[or the Supplemental Provisions to the Project Document]*, the responsibility for the safety and security of each responsible party, subcontractor and sub-recipient and its personnel and property, and of UNDP's property in such responsible party's, subcontractor's and sub-recipient's custody, rests with such responsible party, subcontractor and sub-recipient. To this end, each responsible party, subcontractor and sub-recipient shall:
        i. put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;
        ii. assume all risks and liabilities related to such responsible party's, subcontractor's and sub-recipient's security, and the full implementation of the security plan.

[9] To be used where UNDP is the Implementing Partner

[10] To be used where the UN, a UN fund/programme or a specialized agency is the Implementing Partner

b. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the responsible party's, subcontractor's and sub-recipient's obligations under this Project Document.

c. Each responsible party, subcontractor and sub-recipient will take appropriate steps to prevent misuse of funds, fraud or corruption, by its officials, consultants, subcontractors and sub-recipients in implementing the project or programme or using the UNDP funds. It will ensure that its financial management, anti-corruption and anti-fraud policies are in place and enforced for all funding received from or through UNDP.

d. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to each responsible party, subcontractor and sub-recipient: (a) UNDP Policy on Fraud and other Corrupt Practices and (b) UNDP Office of Audit and Investigations Investigation Guidelines. Each responsible party, subcontractor and sub-recipient agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

e. In the event that an investigation is required, UNDP will conduct investigations relating to any aspect of UNDP programmes and projects. Each responsible party, subcontractor and sub-recipient will provide its full cooperation, including making available personnel, relevant documentation, and granting access to its (and its consultants', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with it to find a solution.

f. Each responsible party, subcontractor and sub-recipient will promptly inform UNDP as the Implementing Partner in case of any incidence of inappropriate use of funds, or credible allegation of fraud or corruption with due confidentiality.

   Where it becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, each responsible party, subcontractor and sub-recipient will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). It will provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

g. Choose one of the three following options:

   Option 1: UNDP will be entitled to a refund from the responsible party, subcontractor or sub-recipient of any funds provided that have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of this Project Document. Such amount may be deducted by UNDP from any payment due to the responsible party, subcontractor or sub-recipient under this or any other agreement. Recovery of such amount by UNDP shall not diminish or curtail any responsible party's, subcontractor's or sub-recipient's obligations under this Project Document.

   Where such funds have not been refunded to UNDP, the responsible party, subcontractor or sub-recipient agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to such responsible party, subcontractor or sub-recipient for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

   Note: The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, subcontractors and sub-recipients.

h. Each contract issued by the responsible party, subcontractor or sub-recipient in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in

contract execution, and that the recipient of funds from it shall cooperate with any and all investigations and post-payment audits.

i. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project or programme, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP.

j. Each responsible party, subcontractor and sub-recipient shall ensure that all of its obligations set forth under this section entitled “Risk Management” are passed on to its subcontractors and sub-recipients and that all the clauses under this section entitled “Risk Management Standard Clauses” are adequately reflected, *mutatis mutandis*, in all its sub-contracts or sub-agreements entered into further to this Project Document.

XI. ANNEXES

1. **Project Quality Assurance Report**

2. **Social and Environmental Screening Template** [English][French][Spanish], including additional Social and Environmental Assessments or Management Plans as relevant.
   (NOTE: The SES Screening is not required for projects in which UNDP is Administrative Agent only and/or projects comprised solely of reports, coordination of events, trainings, workshops, meetings, conferences, preparation of communication materials, strengthening capacities of partners to participate in international negotiations and conferences, partnership coordination and management of networks, or global/regional projects with no country level activities).

3. **Risk Analysis**. Use the standard Risk Log template. Please refer to the Deliverable Description of the Risk Log for instructions

4. **Capacity Assessment:** Results of capacity assessments of Implementing Partner (including HACT Micro Assessment)

5. **Project Board Terms of Reference and TORs of key management positions**

Assumptions 1-5 feddan

Irrigated area	1-5	feddan
Exchange rate	19	SDG/US$
Number of pumps	250	pump sets
VAT	17%	
Insurances	5%	
Module size	260	Wp
Number of modules	12	module
Total power	3.12	kWp
Pumping rate per day	100	M^3/day
Total head	10	meter
Module area (669*1.484 m)	24.82	m2
Installed area	29.784	m2

Item Descriptions & specifications	QTY	Unit	Unit cost	Total
Lorentz PS 21k. Centrifugal pump, C-SG 150-17/4	1	pcs	1,500	1500
Water level sensor for dry run protection + wire	1	pcs	60	60
Lorentz Solar Module LC185-24m, Polycrystalline	12	pcs	128	1536
Support Structure: - Bolted structure for modules mounting, it withstand high wind speed (40 m/s)	1	sets	700	700
Junction box	1	Pcs	250	250
Pump Drop Cable: flexible 4 core cable with 4x6 mm2	25	meter	3	75

Termination Kit for the Drop cable for 8 above	1	Pcs	10	10
Pump Security wire, Rust proof type	25	meter	40	1000
Galvanized steel pipes (Diameter 8 inch, 3 meters long, threaded on both sides)	20	Pcs	45	900
Steel box for controller + pipe fittings + wires	1	set	250	250
Installation materials	1	set	100	100
A. Total pump cost				6381
Insurances				267
Transportation of Goods & Team, accommodation and other expenses				200
Installation fees				500
B. Others costs				967
C. Grand Total pump cost (A+B)				**7348**

*Note: Estimated number of pumps 250*7348$ per pump=1,837,000$*

Assumptions 5-10 feddan

Irrigated area	5-10	feddan
Exchange rate	19	SDG/US$
Number of solar pumps	200	pump sets
VAT	17%	
Insurances	5%	
Module size	260	Wp
Number of modules	20	module
Total power	5.2	kWp
Pumping rate 150 Cubic meters per day	300	M^3/day
Total head	10	meter
Module area (0.55*1.21)	13.31	m2
Installed area	19.965	m2

Item Descriptions & specifications	QTY	Unit	Unit cost	Total
Lorentz PS 21k. Centrifugal pump, C-SG 150-17/4	1	pcs	2,000	2000
Water level sensor for dry run protection + wire	1	pcs	60	60
Lorentz Solar Module LC185-24m, Polycrystalline	20	pcs	128	2560
Support Structure: - Bolted structure for modules mounting, it withstand high wind speed (40 m/s)	1	sets	1,000	1000
Junction box	1	Pcs	250	250
Pump Drop Cable: flexible 4 core cable with 4x6 mm2	25	meter	3	75
Termination Kit for the Drop cable for 8 above	1	Pcs	10	10
Pump Security wire, Rust proof type	25	meter	40	1000
Galvanized steel pipes (Diameter 8 inch, 3 meters long, threaded on both sides)	20	Pcs	45	900
Steel box for controller + pipe fittings + wires	1	set	250	250
Installation materials	1	set	100	100
A. Total pump cost				8205
Insurances				350
Transportation of Goods & Team, accommodation and other expenses				250
Installation fees				600
B. Others costs				1200
C. Grand Total pump cost (A+B)				**9405**

*Note: Estimated number of pumps 200*9405$ per pump=1,881,000$*

Total amount of the pumps **(***1,837,000$+1,881,000$=3,718,000$)*

Average amount of the fund per pump *is (3,718,000/450 pumps=8,262$ per pump)*

Assumption for CO2 emission

The calculation is based on the diesel fuel consumption per day (20 liters for small pump and 30 liters) times the days of irrigation per season (180 days in winter season and 90 days in summer season). Thus, the annual diesel consumption is calculated for each size of pumps as follows:

Items	Diezel Pumps		Total
	1-5 feddan	5-10 feddan	
Fuel consumption liters per days	20	30	
Annual fuel consumption pump size (liter/diesel pump/year)	5,400	8,100	
Number of pumps to be replaced by the project	250	200	450
Annual diesel fuel consumption by all pumps replaced pump sizes (Metric ton/year)	1,350	1,620	2,970
Annual Emission Reduced	4,320	5,184	9,504

Annex 5. Environmental Screening Checklist

Question 1. Related environmental impacts

1-1 Will the project include new construction or enlargement or repair of building?

□ Yes □X No

1-2 Will the project change geographical features or land use?

□ Yes □X No

1-3 Will the project use or develop water resources?

□X Yes □ No

1-4 Will the project produce waste?

□ Yes □X No

Question 2. Requirement of Environmental Impact Assessment (EIA)

2-1 Does the host country have Environmental Impact Assessment (EIA) related laws or guidelines?

□X Yes □ No

2-2 Is EIA required for the project according to the laws or guidelines in the host country?

□ Yes □X No □ Unknown

2-3 If the EIA is required, please mark the corresponding item.

□ Implemented □ On going □ Planning

Question 3. Project's sensitive features

3-1 Does the project come under following sectors?

□ Yes □X No

If yes, please mark the corresponding items.

□ Hydropower, dams and reservoirs □ Urban development

□ Roads, railroads and bridges □ Airports, ports and harbor

□ Water supply, sewage treatment □ Solid waste treatment

□ River/Sand control □ Power transmission and distribution lines

□ Mining development □ Industrial development

□ Forestry □ Fishery □ Tourism

☐ Agriculture (large-scale land-clearing or irrigation)

3-2 Is any of the following area located on or around the project site?

□ Yes □X No

If yes, please mark corresponding items.

□ National park

□ Protected area designated by the government (cultural heritage)

□ Protected area designated by the government (coastal zone, wetlands, reserved area for ethnic or indigenous people)

□ Habitat of valuable species protected by domestic laws or international treaties

□ Virgin forests, tropical forests

□ Buffer zone of protected area

□ Ecological important habitat areas (coral reef, mangrove wetland, tidal flats)

□ Likely salts cumulus or soil erosion areas on a massive scale

□ Remarkable decertification trend areas

□ Archaeological, historical or cultural valuable areas

□ Living areas of ethnic, indigenous people or nomads who have a traditional lifestyle or special socially valuable areas

Annex 6. Gender & Development Screening Checklist

Question1.	**Women/Girls as beneficiaries or participants**
1-1	Does this project include women or girls as a target group as direct or indirect beneficiaries? □ X Yes □ No
1-2	Do women or girls participate in implementing process as one of the decision makers? □ X Yes □ No
1-3	Did this project proposal complete based on the needs assessment of men and women/boys and girls? □ Yes □ X No
Question 2.	**Relevance to women's empowerment**
2-1.	What are the key gender issues in the sector/subsector that are likely to be relevant to this project or program? Gender inequalities in Sudan exists and due to years of conflicts and

	security circumstances, women are often disproportionally affected. By using solar pump small-scale irrigation facilities and providing communities with water efficient technologies and training on improved farming practices will i) improve nutritional status of households, particularly women and children, and increased income by selling produce in local markets; ii) enhance access to water for human and livestock consumption; and iii) increased women's participation in farming activities resulting in enhanced small investments. In addition, capacity building, trainings and ensuring balanced political participation and decision-making of women and men are important component of the project and their ability to use adaptation technologies are essential in ensuring the sustainability of the project interventions. Moreover, building on good practices, ensuring the representation of women in technical committees has been beneficial in enhancing their political participation and decision-making opportunities. In addition, ensuring balanced representation within state-level committees and task forces as well as within the project management team can contribute to ensuring the integration of gender-considerations in all parts of the project management, including in the monitoring and evaluation.
2-2.	Does the proposed project or program have the potential to make a contribution to the promotion of gender equity and/or empowerment of women by providing women's access to and use of opportunities, services, resources, assets, and participation in decision making? □ X Yes (→2-2-1) □ No
2-2-1	If yes, what measures are included in the project design to promote gender equality and women's empowerment— □ Gender action plan □ X Other actions or measures □ No action or measure
2-3	Can the proposed project have an adverse impact on women and/or girl or widen gender inequality? □ Yes □ X No
2-4	Indicate the intended gender mainstreaming category*: □ GEN (gender equity) □ X EGM (effective gender mainstreaming) □ SGE (some gender elements) □ NGE (no gender elements)

○ **GEN : Gender Equity**

- A project is assigned GEN, if the project outcome directly addresses gender equality and/or women's empowerment by narrowing gender disparities through access to social services(e.g. education, health, and water supply/sanitation); and/or economic and financial resources and opportunities (e.g. employment opportunities, financial services, land, and markets), and/or basic rural and urban infrastructure(e.g. rural electrification, rural roads, pro-poor energy distribution, and urban services for the poor); and/or enhancing voices and rights(e.g. decision making process and structures, political empowerment, and grievance mechanisms); and,
- the outcome statement of the project design and monitoring framework(DMF) explicitly mentions gender equality and women's empowerment and/or, the outcome performance indicators include gender indicators.

○ **EGM : Effective Gender Mainstreaming**

- A project is assigned EGM, if the project outcome is not gender equality or women's empowerment, but project outputs are designed to directly improve women's access to social services, and/or economic and financial resources and opportunities, and/or basic

rural and urban infrastructure, and/or enhancing voices and rights, which contribute to gender equality and women's empowerment.

○ **SGE : Some Gender Elements**

- A project is assigned SGE, if it meets either of the following:

(i) by its nature it is likely to directly improve women's access to social services; and/or economic and financial resources and opportunities, and/or basic rural and urban infrastructure, and/or enhance their voices and rights(e.g. education, health, rural development, microfinance, water supply and sanitation, food security, and emergency food and rehabilitation assistance), but that included little, if any gender analysis and few or no specific design features; and did not meet the EGM criteria, or

(ii) is unlikely to directly improve women's access to social, economic or financial resources or opportunities, but significant efforts were made during project preparation to identify potential positive and negative impacts on women. Some gender features are included to enhance benefits to women(e.g. targets for employment of women in project construction work, provision of equal pay for equal work, information campaigns on HIV/AIDS risk, gender training of executing/implementing agencies, and adherence to core labor standards, esp. child labor); and where resettlement is involved includes attention to women in the mitigation/resettlement plans(such as compensation payments to both men and women, joint-ownership of replacement land/housing, restoration of livelihood initiatives for women, and so forth).

○ **NGE : No Gender Elements**

- A project is assigned NGE, when it does not include any gender design features(e.g. no indicator or goal for gender equality, no women's participation or empowerment, no resource or no opportunity for women, etc.).

**Sourced from ADB (2012) Guidelines for Gender Mainstreaming Categories of ADB Projects*

Cabinet Decision: Tax exemptions for solar irrigation system (pdf file attached)

Ministry of Finance and Economic Planning: Approval for Soalr PV fund establishment (pdf file attached)

[i] UNESCO (2009), Electricity Access Rates.
[ii] Arab Union of Electricity (2012), Statistical Bulletin.
[iii] RCREEE (2013), Arab Future Energy Index.

Annex (7)

The National Photovoltaics (PV) fund

Introduction:

The National PV Fund is a fund developed by the Government of Sudan, with the aim of supporting the promotion and deployment of solar energy to operate water pumps in under irrigation areas in Sudan. The Fund represents an innovative finance mechanism that combines domestic and international economic resources from public and private sectors and connects them directly to rural communities in need across the whole country.

Institutional structure
According to the Sudan's Cabinet approval, resolutions number 58/2017 – Fifth: the axis of irrigation, the current situation and the future vision #1), the Government of Sudan has endorsed the exemption of all solar powered irrigation equipment from customs and other fees. At the same time, the Ministry of Finance and Economic Planning has approved the establishment of the National PV Fund.

Other Sudanese public institutions are also involved in the design and implementation of the National PV Fund, such as the Ministry of Water Resources and Electricity (MWRIE), Ministry of Agriculture, Central Bank of Sudan, in coordination with state level institutions such as State Ministries of Agriculture or Finance.

Financial mechanism and partnerships
The Fund is populated with resources from a range of domestic and foreign organizations, which currently includes Treasury of Sudan, Sudan's banking sector, private sector technology suppliers, international finance mechanisms such as grant funds provided by GEF and bilateral partners such as Korea or China. Potential future partners include International Finance Institutions and the engagement of different areas in the private sector.

Deposited funds will be accessible to farmers through loans. The subsidy scheme for the purchase and installation of the water pump equipment is as follows: 50% is

subsidized by the Ministry of Finance, 25% is subsidized by the State Government, 12.5% is directly subsidized by the specific project (with contributions from external partners), and the farmer is responsible for the remaining 12.5%, which is obtained through the loan. Re-payment of the loan takes place every 12 months, for a period of 4-7 years.

The Fund is administered by a Bank Consortium, which follows financing policies, regulations and measures of credit eligibility established by the Bank of Sudan. The Consortium collects funds from each of the participating banks, which will then lend the fund through their branches according to the stipulated Fund rules.

Implementation
The implementation of the National PV Fund is orchestrated throughout an institutional mechanism that occurs at two levels, national and state, and eventually linked to the community level formed by rural farmers (see Figure 1).

The overall coordination of the Fund is performed through a Steering Committee, headed by the MWRE, that acts as a board of directors. Working plans endorsed by the Steering Committee are executed by the Project Management team, which counts with legal, technical, administrative and financial advisors, who are responsible for the implementation and installation of the solar pumps. The Steering Committee and the Project Management will work in collaboration with the Ministry of Finance and the Central Bank of Sudan.

At the State level, the State Ministry of Agriculture acts as a link between the Steering Committee in Khartoum and regional State authorities. It supervises, manages and monitors the installation and after sale services of the solar pumps, through the support of a Technical Committee supervised by the national Project Management team. In coordination with the Bank Consortium, assists on loan process and selection of loan-granted farmers.

The supplying companies, at either national or state levels, provide and install the pumps, including training and after sale services as needed. Contracted by the national Project Management, it is a requirement that every component is 25-year guaranteed.

Lastly, farmers apply to the Technical Committee and to one of the bank branches to receive technical and loan approval, respectively. If successful, farmers will receive equipment and training from the supply companies to operate the solar pumps and re-pay the loan annually as stipulated.

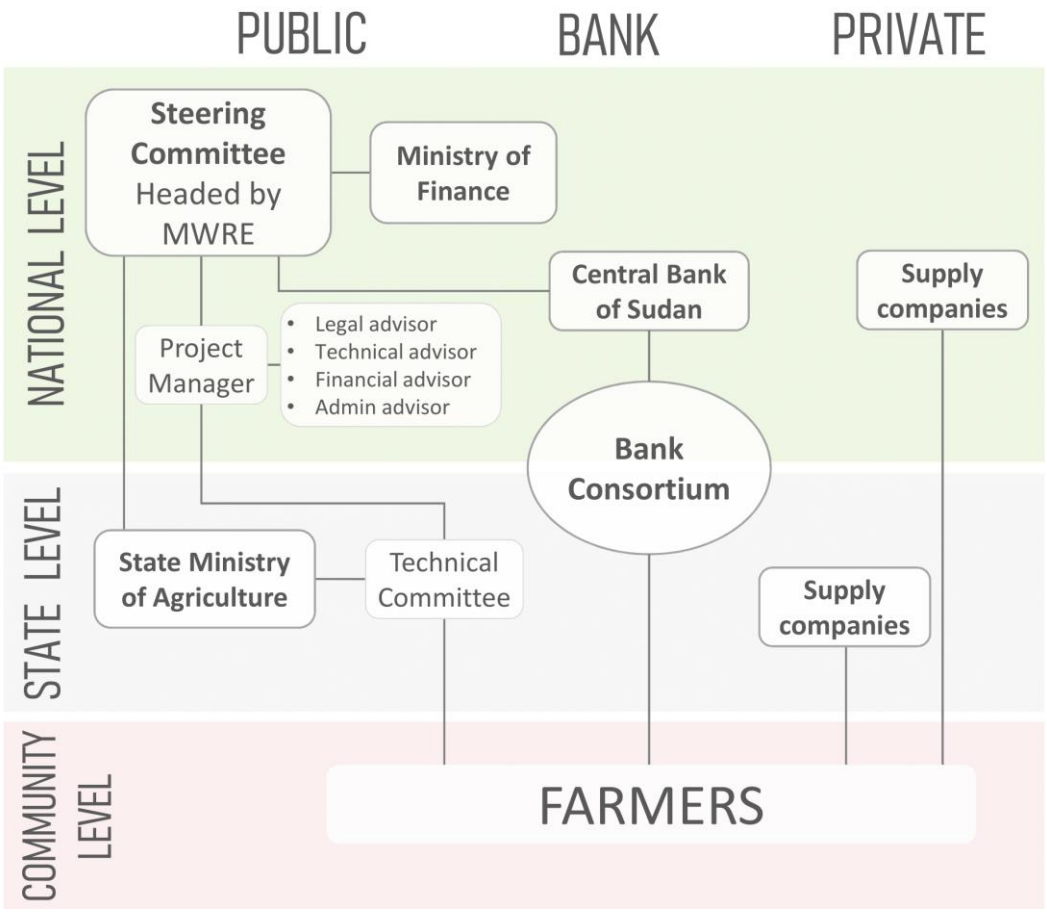


Figure 1. Institutional structure for the implementation of the National PV Fund.